

Lecture 5

Nudges

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PS4168: Economic Psychology

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Nudge (Thaler & Sunstein, 2008)

Richard H. Thaler
Cass R. Sunstein

Nudge



Improving Decisions
About Health, Wealth,
and Happiness

Overview

- Background
 - Recap on heuristics/biases
 - Brief overview of dual-systems
- Nudges: Examples and Definition
- Nudges and Theory
- In-class Activity

Recap on last week!

- In Groups:
 - List 7 biases
 - and be ready to explain 3
 - List 5 Heuristics
 - and be ready to explain 2

Background

Background

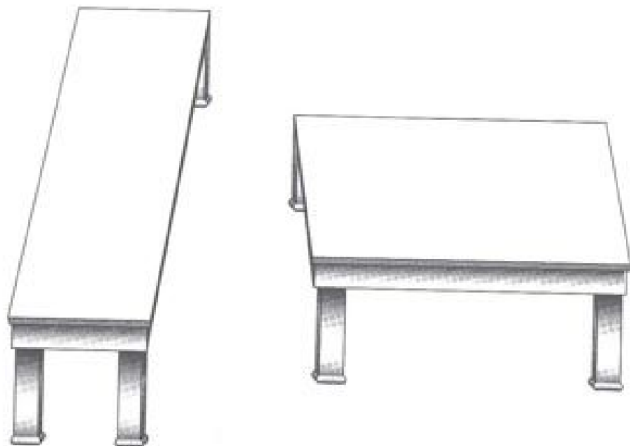
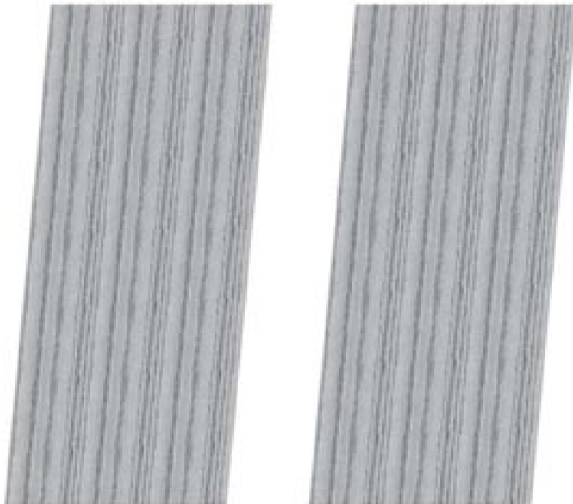


Figure 2: tables

Background



From Biases to Nudges

- Recall that a bias is:
 - “An inclination towards a position or conclusion” (Reber, 2001, p. 88)
 - A tendency to:
 - act in a particular way
 - make judgements in a particular way
 - An error in reasoning (Eysenck & Keane, 2005, p. 512)
- In some cases a bias is the “failure” of a heuristic judgement
 - A heuristic applied in the wrong context

From Biases to Nudges

- Heuristics and biases linked by similarities in underlying cognitive mechanisms
- Errors in applying heuristics manifest as biases
 - These biases led to the discovery of specific heuristics
 - (e.g., representativeness / availability)
- Study of heuristics and biases informed our understanding of how people think
 - Dual-process/Dual-systems Theories

Dual-processes / Dual-systems

Dual-processes / Dual-systems

Two cognitive systems

Automatic System

Uncontrolled

Effortless

Associative

Fast

Unconscious

Skilled

Reflective System

Controlled

Effortful

Deductive

Slow

Self-aware

Rule-following

Figure 4: tables

(taken from Thaler & Sunstein, 2008, p. 20)

Dual-processes / Dual-systems



Figure 5: “Homer and Spock”

Dual-processes / Dual-systems



Figure 6: "Homer and Spock"

Harnessing our Biases

- “System 1” → Heuristics
 - (mis-applied) Heuristics → Biases
 - Biases → poor choices (sometimes)
- **But** understanding biases → *Development of an Intervention*
 - *Intervention* → Good choices
- Specific type of intervention:
 - Nudge

What is a Nudge?

What is a Nudge?



Figure 7: tables

Fly in the Urinal

- Schiphol Airport in Amsterdam
- Fly etched into each urinal
- “Spillage” reduced by 80% (Thaler & Sunstein, 2008)
- Variations in other airports
 - A soccer goal equipped with a ball during the World Cup
- “Fly in the urinal has become the perfect exemplar of a nudge”(Thaler, 2015, p. 252)

Defining Nudges

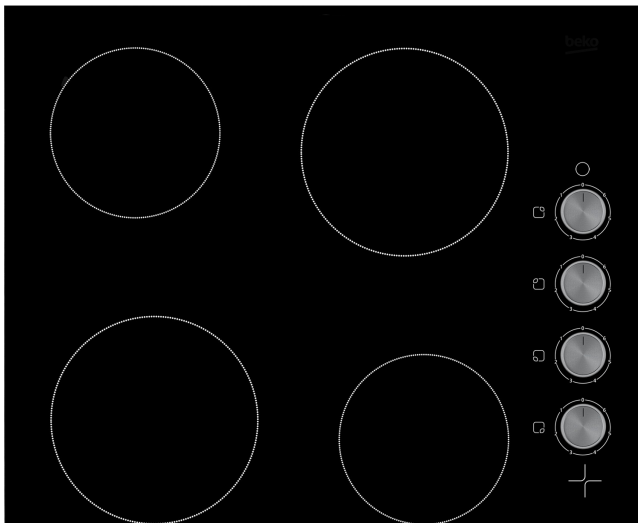
- “A nudge is some small feature in the environment that attracts our attention and influences behavior” (Thaler, 2015, p. 252)
- “any aspect of the choice architecture that alters people’s behavior in a predictable way
 - without forbidding any options
 - or significantly changing their economic incentives” (Thaler & Sunstein, 2008, p. 6)
- Must be easy and cheap to avoid
- Nudges are not mandates:
 - Putting the fruit at eye level counts as a nudge
 - Banning junk food does not (Thaler & Sunstein, 2008, p. 6)

Defining Nudges and “Better Choices”

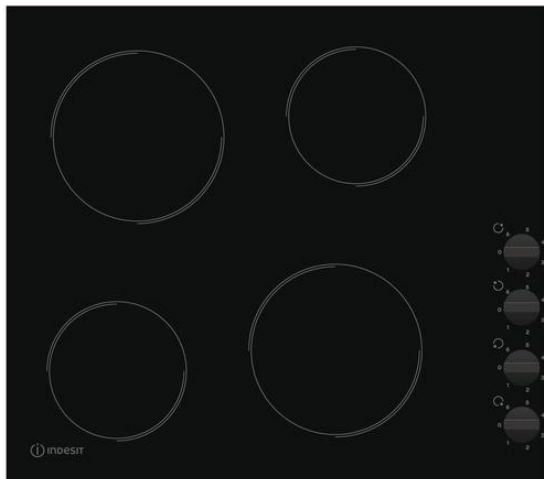
- Libertarian Paternalism
- Paternalism but “free to choose”
- Aim of Libertarian Paternalism is to:
 - “influence choices in a way that will make choosers better off,
as judged by themselves”

(Thaler, 2015, p. 251; Thaler & Sunstein, 2003; 2008, p. 5)

Hobs



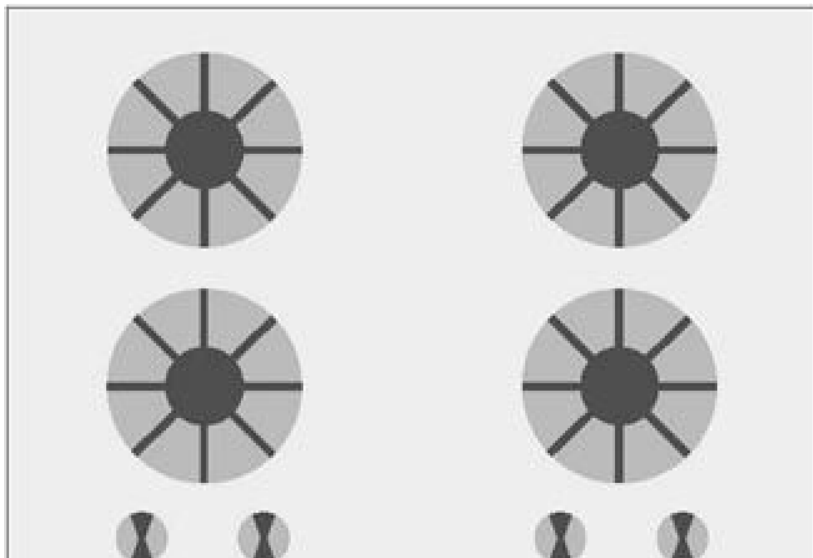
Hobs



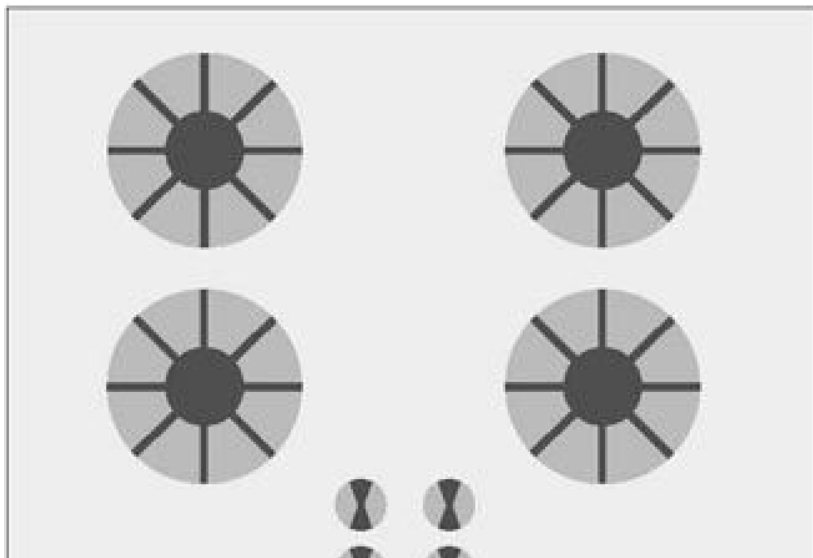
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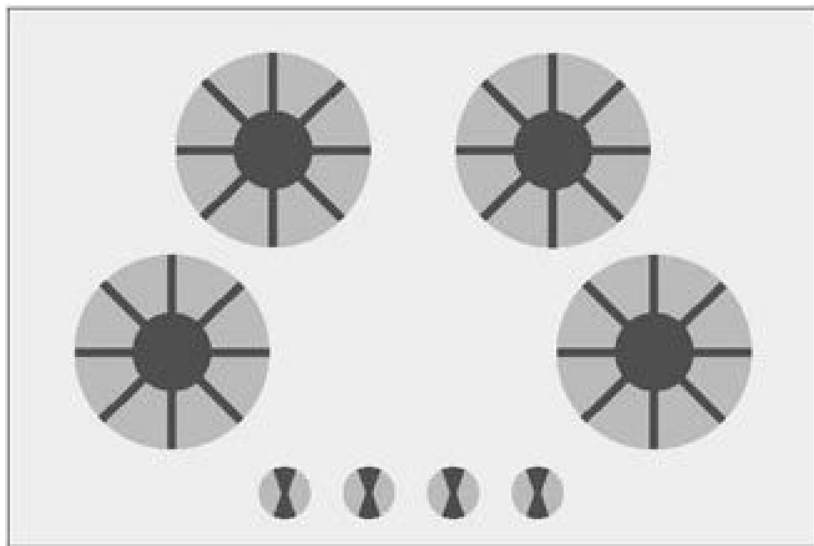
Arrangement of Knobs on a Cooker



Arrangement of Knobs on a Cooker



Arrangement of Knobs on a Cooker



Donating organs

Country	Rate of Donation
USA	28%
France	99.9%
Britain	17%
Hungary	99.9%

Donating organs

Default Rule	Country	Rate of Donation
Explicit Consent	USA	28%
Presumed Consent	France	99.9%
Explicit Consent	Britain	17%
Presumed Consent	Hungary	99.9%

- Opt-in vs opt-out (Gigerenzer, 2008)

Maintaining Standards



Figure 14: tables

Checklists

- Slightly more invasive than other nudges (may not even count as a nudge)
- Hospitals introduced a surgical checklist during major operations
- 8 Cities Globally: Ifakara (Tanzania), Manila (Philippines), New Delhi (India), Amman (Jordan), Seattle (United States of America), Toronto (Canada), London (United Kingdom) and Auckland (New Zealand)
- Data were from 7,688 patients
 - 3,733 before and 3,955 after checklist was introduced
- Incidences of major complications fell by $\frac{1}{3}$ (11% to 7%)
- 40% Drop in in-patient deaths (1.5% to 0.8%) (Epstein, 2009; Gawande, 2010; Haynes et al., 2009; Weiser et al., 2010)

Nudges and Theory

Nudges and Theory

- Nudges are designed to exploit features of our decision making
 - e.g., appealing to loss aversion in persuasion
- The remaining nudges described will be linked with specific theoretical underpinnings
 - Exploiting specific biases
 - Tailored for specific heuristics
 - Appealing to other features of decision making

Activity

- In groups, Identify the underlying theory behind the nudges discussed
 - What heuristics/biases do they harness?
 - Is there any other concept covered that might be relevant?
(some may not have been covered yet, but you can guess)
- Fly; Cooker; Donating organs; Checklist

Activity (possible answers)

- Fly (focusing attention: combating “mindless”/“autopilot” behaviour)
- Cooker (path of least resistance / cognitive ease)
- Donating organs (Default heuristic)
- Checklist (combating “mindless”/“autopilot” choosing)

Principles relevant to Nudges

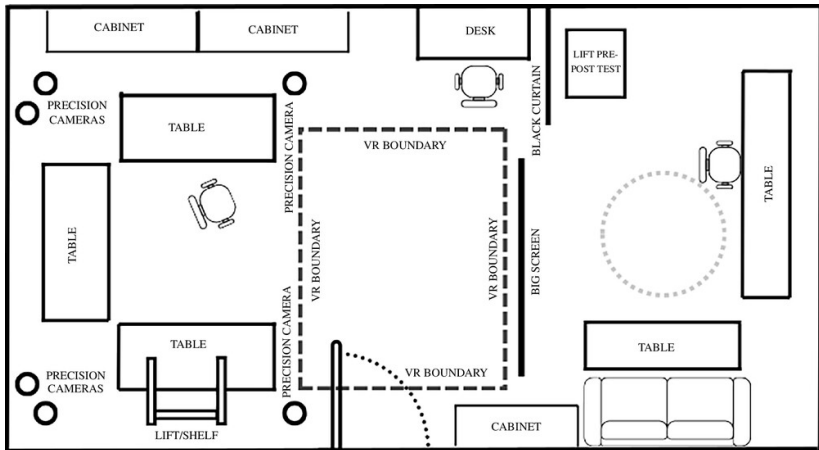
- Choice architecture (Thaler, Sunstein, & Balz, 2012)
- Path of least resistance
- Heuristics: Anchoring / Availability / Representativeness
- Biases: Optimism / over-confidence / Status-quo bias
- Loss aversion
- Emotion / affect / mood
 - *Hot vs cold* decisions
- “mindless” or “autopilot” choosing
- (Mental Accounting)

Choice Architecture

- **Choice architecture** is the “context in which people make decisions” (Thaler & Sunstein, 2008, p. 3)
- Everything matters
- No such thing as “neutral design”

(Thaler et al., 2012)

Choice Architecture?



Behaviour settings (Di Rienzo, Myin, & van Dijk, 2024; see also McGann, Lucas, McHugh, & Barrett, 2024)

Choice Architecture

- 6 principles of good choice architecture
 - iNcentives
 - Understand mappings
 - Defaults
 - Give feedback
 - Expect error
 - Structure complex choices

(Thaler et al., 2012)

Path of Least Resistance (Defaults)

- Easier choice options are preferable to more difficult options
- Consider the placing of items near a till in a shop
- Sweets and crisps could be replaced with healthy snacks/fruit
 - This would lead to people choosing healthy snacks more often
- Too many choices makes decisions more difficult

Heuristics: Anchoring

- Recall Anchoring (heuristics lecture)
- Taking information salient in the environment and using it to *anchor* your decisions
- e.g., rigged wheel of fortune: 10 or 65
 - Is the percentage of African nations among UN members larger or smaller than the number you just wrote?
 - What is your best guess of the percentage of African nations in the UN?
- Mean response 25% and 45% (depending on 10 or 65)
 - In groups/individually:
 - Identify potential applications of anchoring (1-2 mins)

Heuristics: Anchoring

- Taking information salient in the environment and using it to *anchor* your decisions
- Rigged wheel of fortune example (10 or 65)
 - Is the percentage of African nations among UN members larger or smaller than the number you just wrote?
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Heuristics: Anchoring Applications

- Charities looking for donations:
 - Suggested amounts serve as an anchor
 - €100, €250, €1,000, €5,000
 - vs €50, €75, €100, and €150
- The more you ask for the more you get
 - e.g., court cases for compensation/damages

Heuristics: Availability

- the process of judging frequency by “the ease with which instances come to mind”(Kahneman, 2011, p. 128)
- Remind people of a salient (negative) incident and they will over-estimate the risk
 - and you can sell more insurance policies

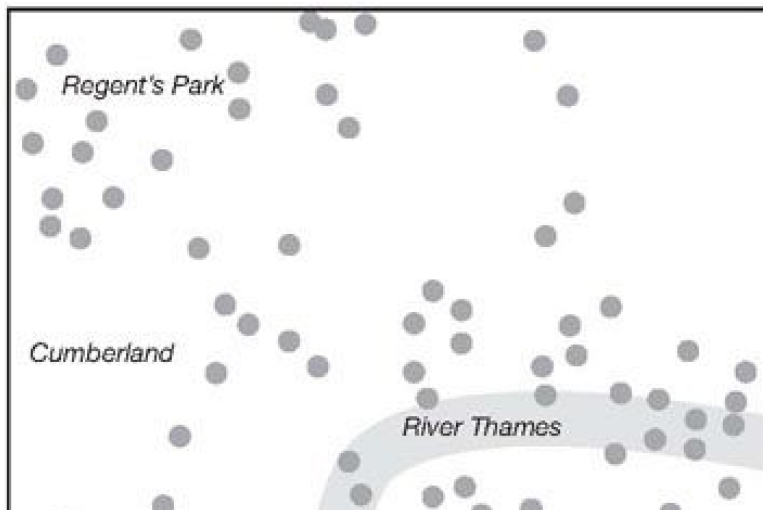
Heuristics: Combatting Availability

- Availability has been blamed for alcohol misuse
 - “everyone does it”
 - leading to more people drinking
- Large scale educational campaign in Montana stressed that majority of citizens do not drink
 - *“Most (81 percent) of Montana college students have four or fewer alcoholic drinks each week.”*
 - *“Most (70 percent) of Montana teens are tobacco free.”*
- The strategy has produced big improvements in the accuracy of social perceptions
 - also statistically significant decreases in smoking(Thaler & Sunstein, 2008, p. 68)

Heuristics: Representativeness

- judging a situation based on how similar the prospects are to the prototypes the person holds in his or her mind.

Map of London: V-1 rocket strikes (WWII)



Heuristics: Representativeness

- Seeing patterns when they are not there
- bombings appear to be around the Thames, and in the north west (German spies in the blank spaces?)
 - but they are random
- Illusion of a “Hot hand” or a “Streak” in basketball (Gilovich, Vallone, & Tversky, 1985; Thaler & Sunstein, 2008)

Optimism

- Optimism bias / over-confidence bias
 - lead to the planning fallacy
- Activity:
 - Identify strategies that may combat the planning fallacy
 - does any of your strategies count as a *nudge*

Status-quo Bias

- “strong tendency to go along with the status-quo or default option” (Thaler & Sunstein, 2008, p. 8)
- New cell phone: default options
- Private companies or public officials think that one policy produces better outcomes, they can greatly influence the outcome by choosing it as the default (Thaler & Sunstein, 2008, p. 8)
- Saving money

Loss Aversion

- Advertising framed in terms of losses
- e.g., “drink more water”:
 - “you sweat in the heat: you lose water.”
- Any other examples?

Emotions / “hot” vs “cold”

- In a few weeks we'll learn that we are very bad at predicting how we will feel “in the moment”
 - Affective forecasting error
- How do we resist the urge to:
 - snack?
- Problems with smoking/alcohol
- Safe sex
- Excessive borrowing/Gambling?

Emotions / “hot” vs “cold”

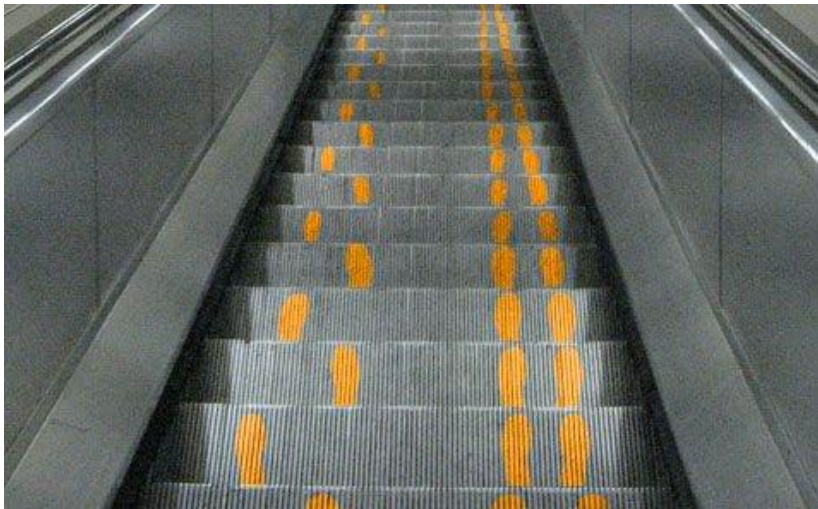
- Alter the “choice architecture”
 - Don't keep snacks in the fridge
 - Dinner party bowl of nuts (Thaler & Sunstein, 2008, p. 40)
 - Don't go shopping on an empty stomach

“mindless” or “autopilot” choosing

- Accidentally driving towards work (on a Saturday)
- Stale popcorn
- Re-filling soup bowls (Thaler & Sunstein, 2008, p. 43)

Other Nudges

Other Nudges



Other Nudges



Other Nudges



Other Nudges



Other Nudges



Further Reading

- Thaler, R. H., Sunstein, C. R., & Balz, J. P. (2012). Choice Architecture. In E. Shafir (Ed.), *The Behavioral Foundations of Public Policy*. Rochester, NY: Social Science Research Network. (uploaded to Brightspace)
- Thaler, R. H., & Sunstein, C. R. (2008). *Nudge: improving decisions about health, wealth, and happiness*. New Haven: Yale University Press.
- Thaler, R. H., & Sunstein, C. R. (2003). Libertarian Paternalism. *American Economic Review*, 93(2), 175–179. (uploaded to Brightspace)
- Thaler, R. H. (2015). *Misbehaving: The Making of Behavioral Economics*. W.W. Norton.

In-class Activity

Nudging Ethics and Applications

- In groups:
 - Discuss the potential ethical implications of nudging
 - Identify potential candidates for a *novel* nudge (7-10 mins)

Avoiding the Ethical Problems

- Aim of Libertarian Paternalism is to:
 - “influence choices in a way that will make choosers better off, ***as judged by themselves***” (Thaler, 2015, p. 251; Thaler & Sunstein, 2003; 2008, p. 5)

References

References

- Di Rienzo, G., Myin, E., & van Dijk, L. (2024). Navigating the normativity of behaviour settings: An observational case study. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 379(1910), 20230295.
<https://doi.org/10.1098/rstb.2023.0295>
- Epstein, D. (2009). WHO | Checklist helps reduce surgical complications, deaths. *World Health Organisation*.
http://www.who.int/mediacentre/news/releases/2009/safe_surgery_
- Eysenck, M. W., & Keane, M. T. (2005). *Cognitive psychology: A student's handbook*. Hove [England]; New York: Psychology Press.
- Gawande, A. (2010). *Checklist Manifesto, The (HB)*. Penguin Books India.
- Gigerenzer, G. (2008). Moral Intuition = Fast and Frugal